Insertion Sort :

Insertion sort algorithm arranges a list of elements in a particular order. In insertion sort algorithm, every iteration moves an element from unsorted portion to sorted portion until all the elements are sorted in the list.

Step by Step Process

The insertion sort algorithm is performed using the following steps...

- Step 1 - Assume that first element in the list is in sorted portion and all the remaining elements are in unsorted portion.
- **Step 2:** Take first element from the unsorted portion and insert that element into the sorted portion in the order specified.
- **Step 3:** Repeat the above process until all the elements from the unsorted portion are moved into the sorted portion.

Following is the sample code for insertion sort...

```
//Insertion sort logic
for i = 1 to size-1 {
   temp = list[i];
   j = i-1;
   while ((temp < list[j]) && (j > 0)) {
      list[j] = list[j-1];
      j = j - 1;
   }
   list[j] = temp;
}
```

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Consider the following unsorted list of elements...



Asume that sorted portion of the list empty and all elements in the list are in unsorted portion of the list as shown in the figure below...



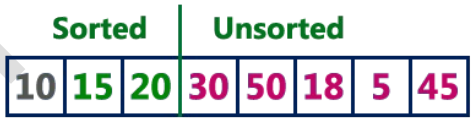
Move the first element 15 from unsorted portion to sorted portion of the list.



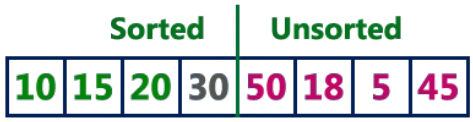
To move element 20 from unsorted to sorted portion, Compare 20 with 15 and insert it at correct position



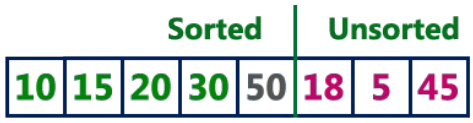
To move element 10 from unsorted to sorted portion, Compare 10 with 20 and it is smaller so swap. Then compare 10 with 15 again smaller swap. And 10 is insert at its correct position in sorted portion of the list.



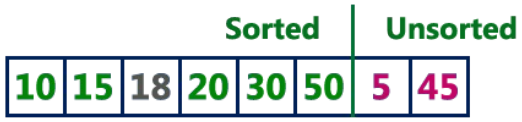
To move element 30 from unsorted to sorted portion, Compare 30 with 20, 15 and 10. And it is larger than all these so 30 is directly inserted at last position in sorted portion of the list.



To move element 50 from unsorted to sorted portion, Compare 50 with 30, 20, 15 and 10. And it is larger than all these so 50 is directly inserted at last position in sorted portion of the list.



To move element 18 from unsorted to sorted portion, Compare 18 with 30, 20 and 15. Since 18 is larger than 15, move 20, 30 and 50 one position to the right in the list and insert 18 after 15 in the sorted portion.



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Complexity of the Insertion Sort Algorithm

To sort an unsorted list with **'n'** number of elements, we need to make **(1+2+3+.....+n-1) = (n (n-1))/2** number of comparisions in the worst case. If the list is already sorted then it requires **'n'** number of comparisions.

Worst Case : O(n^2)
Best Case : Ω(n)
Average Case : Θ(n^2)

Space Complexity : O(1)
Inplace Sorting : Yes
Stable : Yes

Note :

Reference Books : Taken contents and diagrams from various websites.

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